

EE698G: Circuit design for frequency and phase synthesis
Design project #1

Design and Implementation of a 500MHz – 550MHz
Delay-Locked Loop (DLL)

Roll No.	Target Frequency Range	Technology
251040079	500MHz – 550MHz	gpdk 180nm

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ROLL NUMBER : 251040079

Specifications and Requirements

Frequency Range	500 MHz (T = 2.0 ns) to 550 MHz (T = 1.818 ns)
Control Voltage (Vc) Range	0.4V to 1.4V
Supply Voltage (Vdd)	1.8V (Nominal), 1.71V (Min), 1.89V (Max)
Temperature Range	0 ⁰ C to 70 ⁰ C
Static Phase Error (SPO)	less than 25ps

1. Design of voltage-controlled delay line

1.1 VCDL Architecture and Schematic

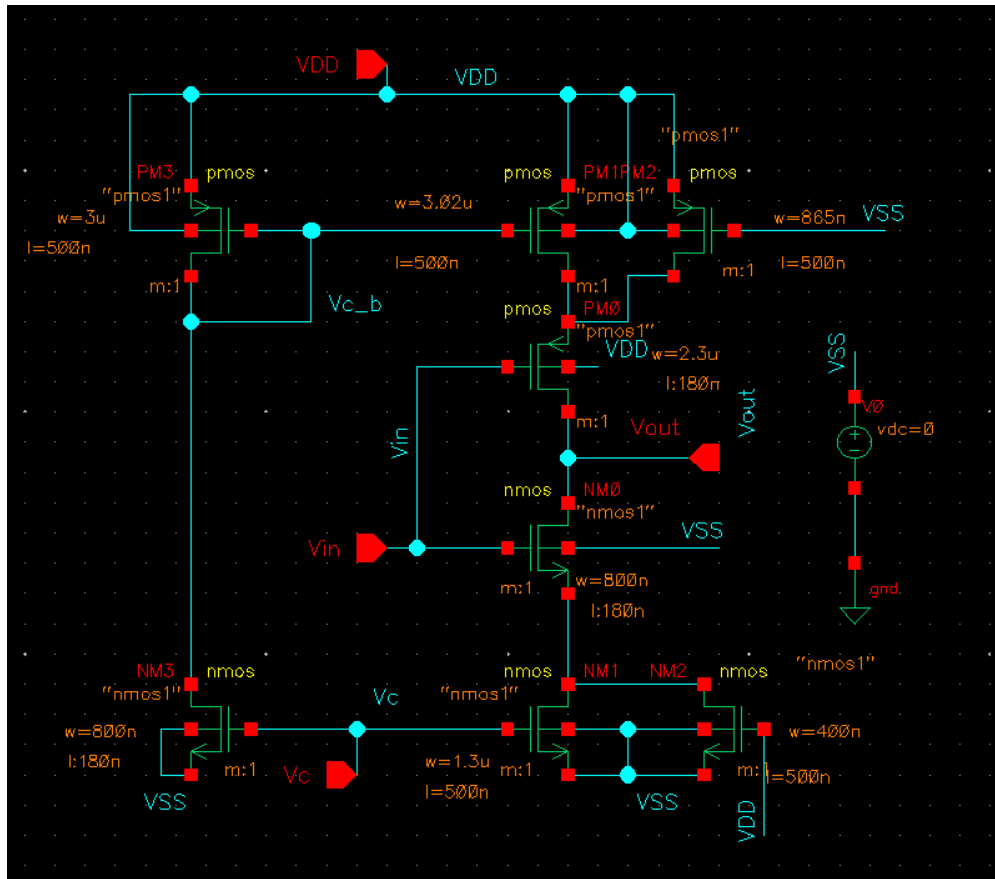


Figure 1: Transistor level current-starved inverter schematic

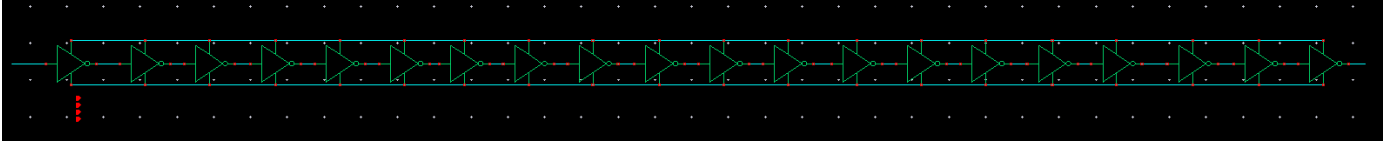
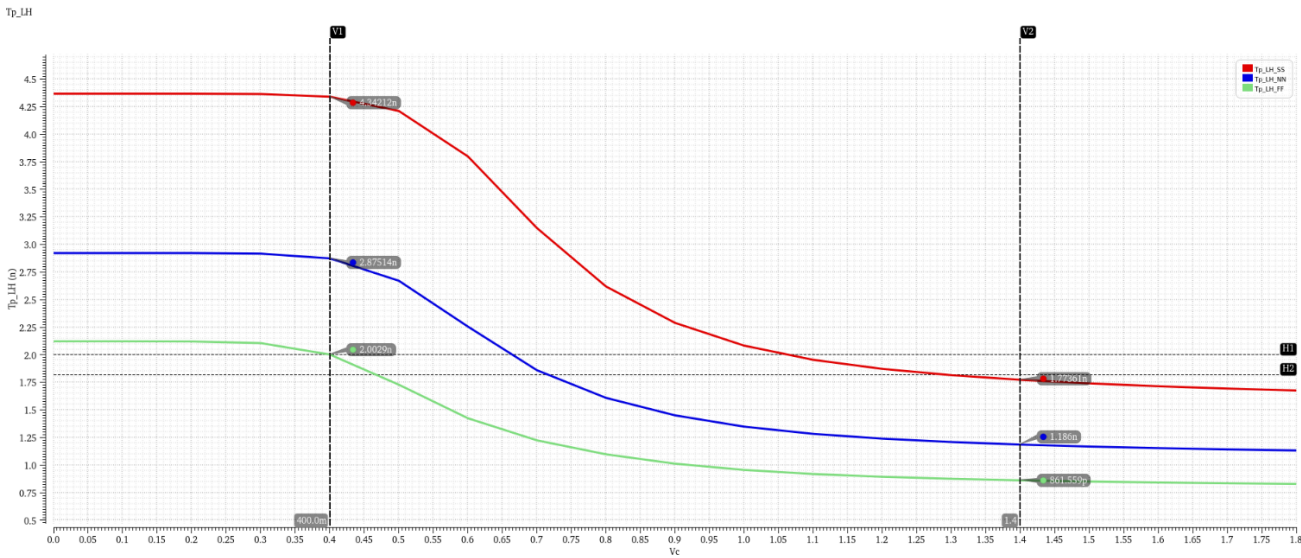
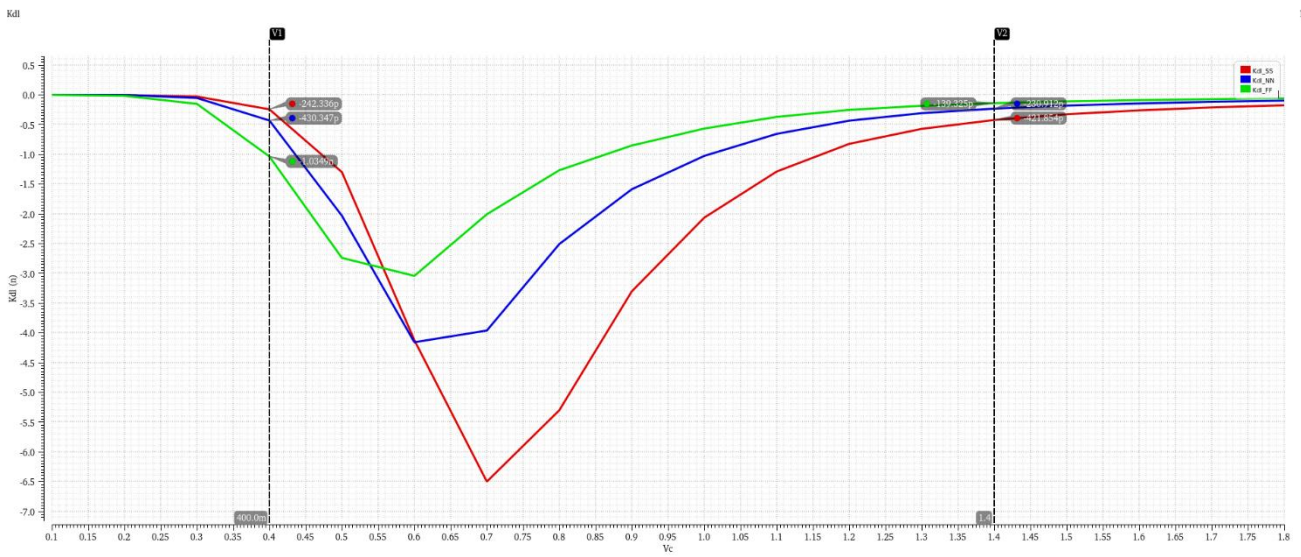


Figure 2: Complete VCDL schematic with a 20-stage current-starved inverter chain.

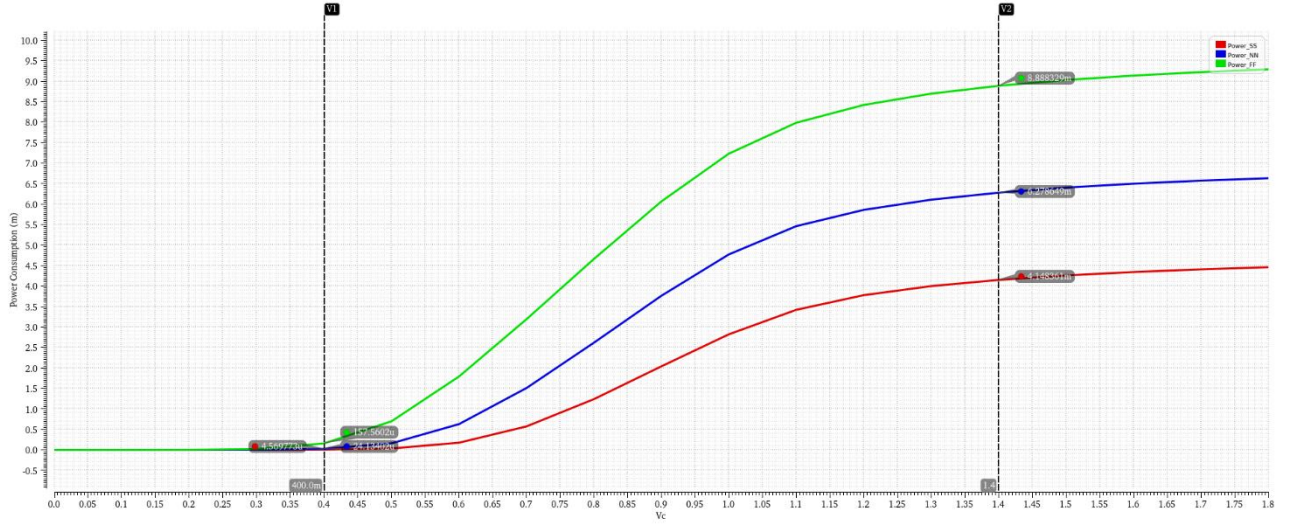
1.2 VCDL Characterization



Plot 1: t_{pLH} vs V_C for VCDL



Plot 2: K_{dl} vs V_C for VCDL



Plot 3: P_d vs V_C for VCDL

2. Design of PFD and charge pump

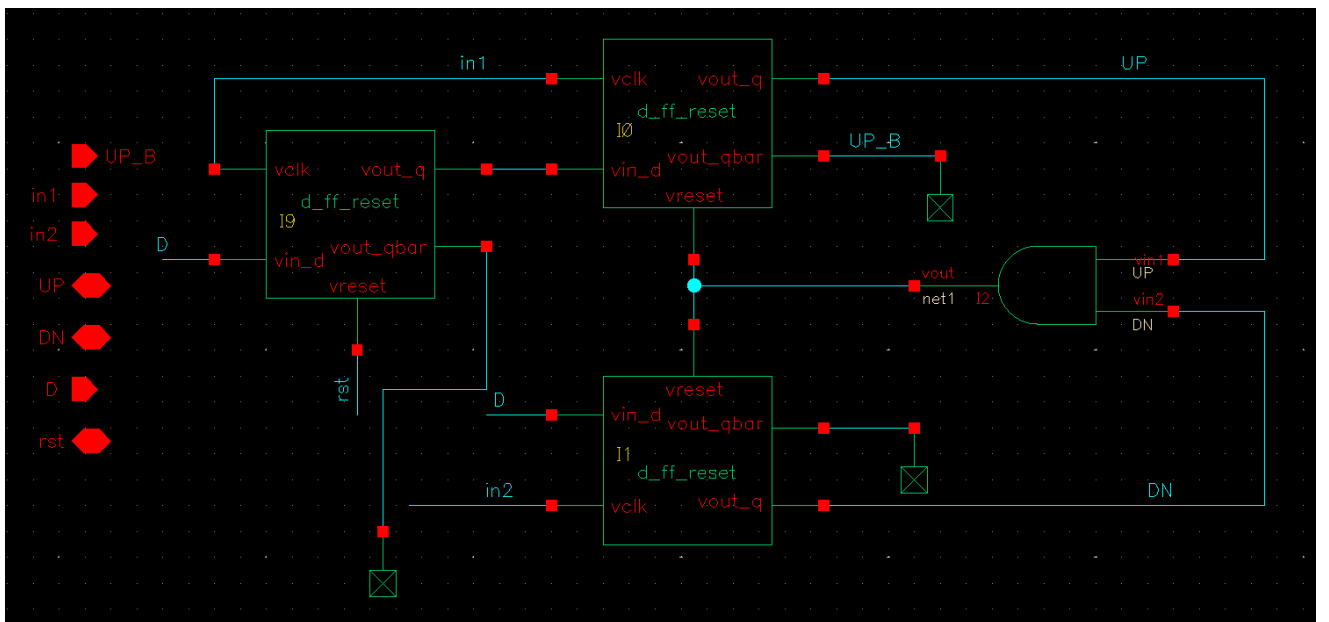


Figure 3: Schematic of PFD using three D-Flip-Flops and an AND gate

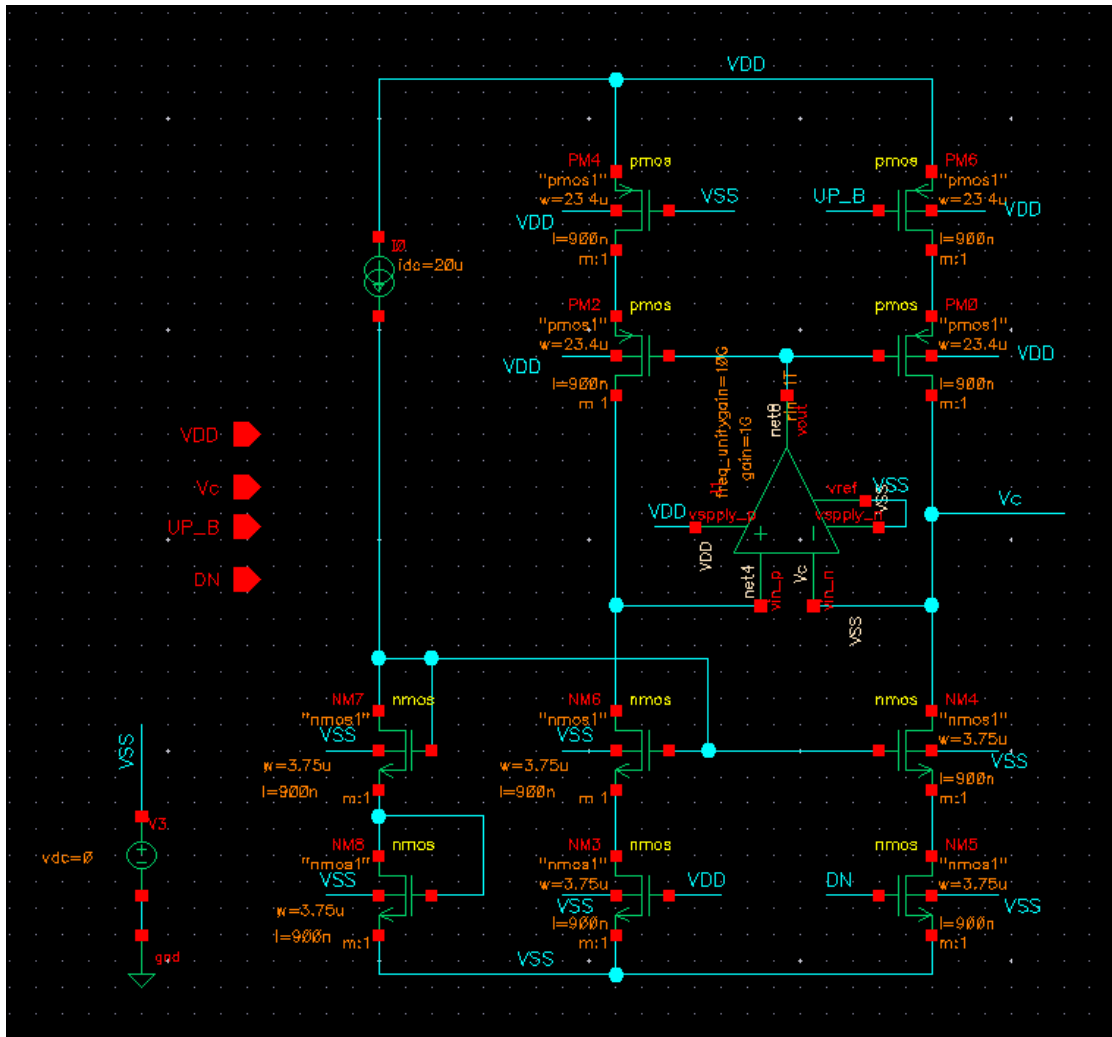
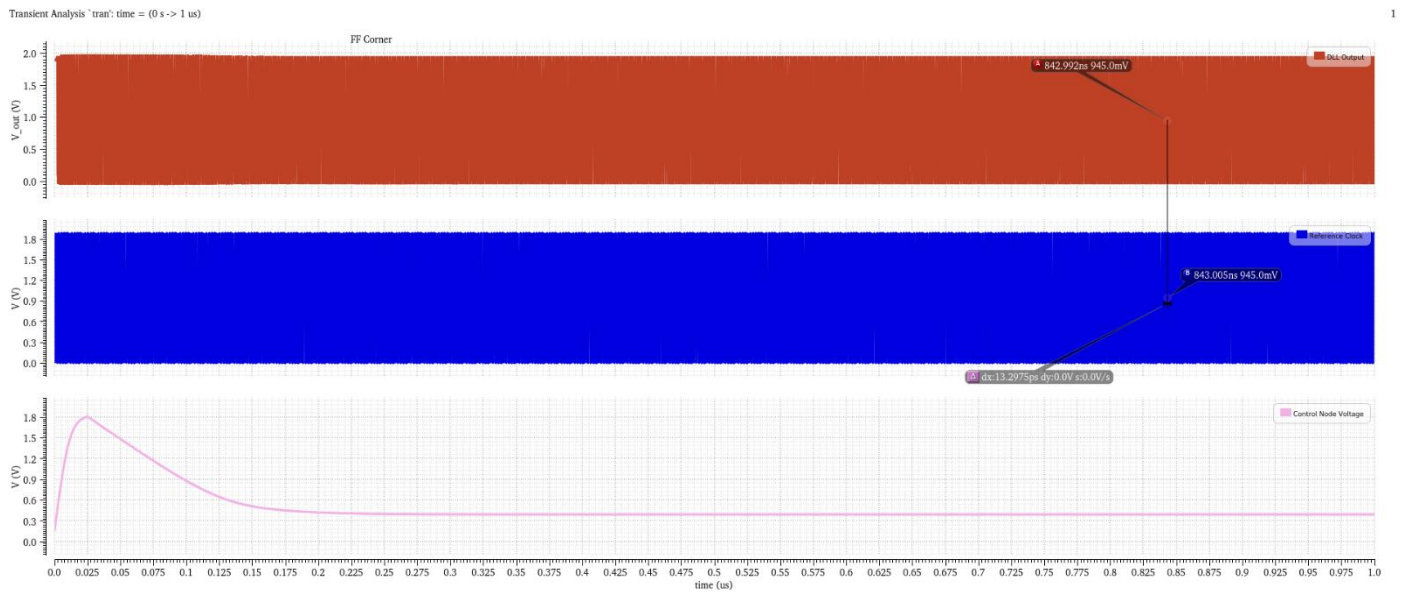
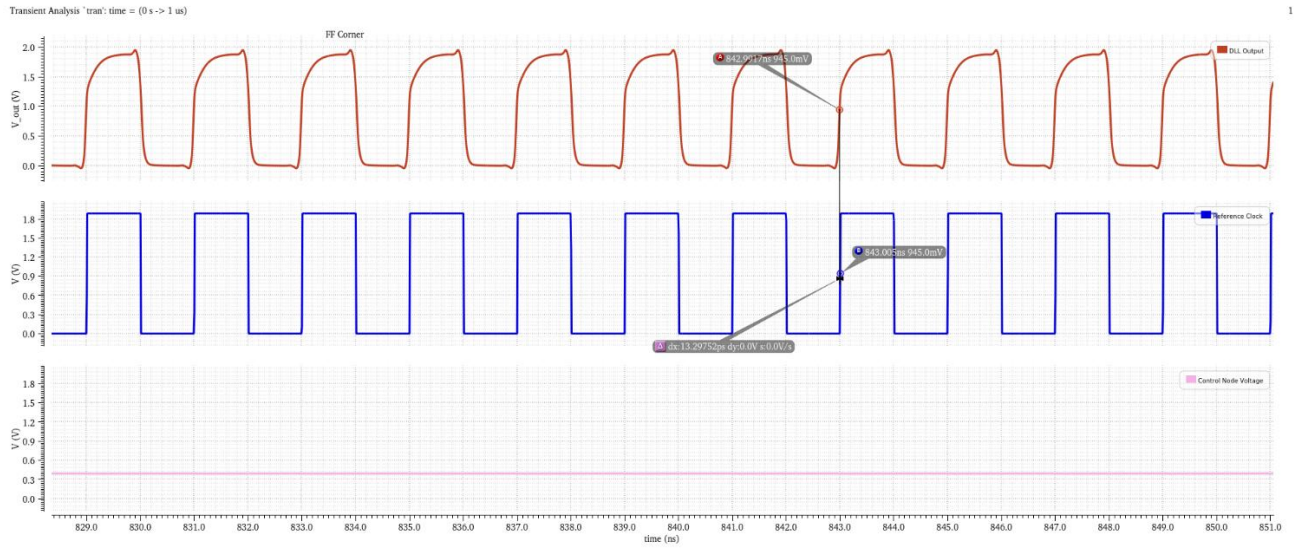


Figure 4: Schematic of Charge Pump uses 20uA current Source

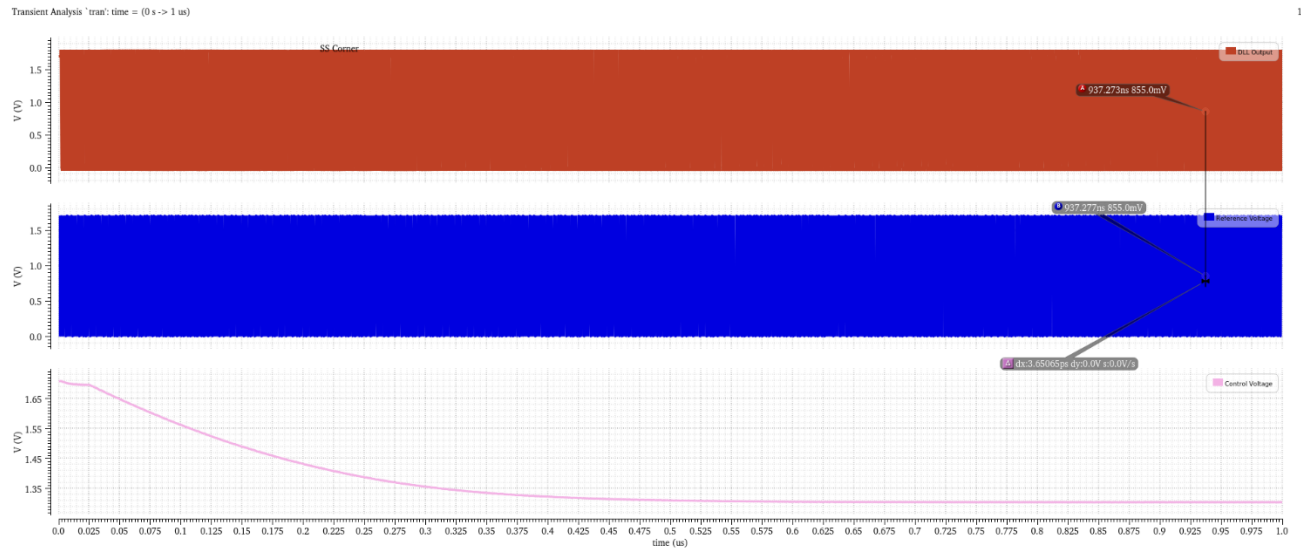
3. Overall DLL Performance



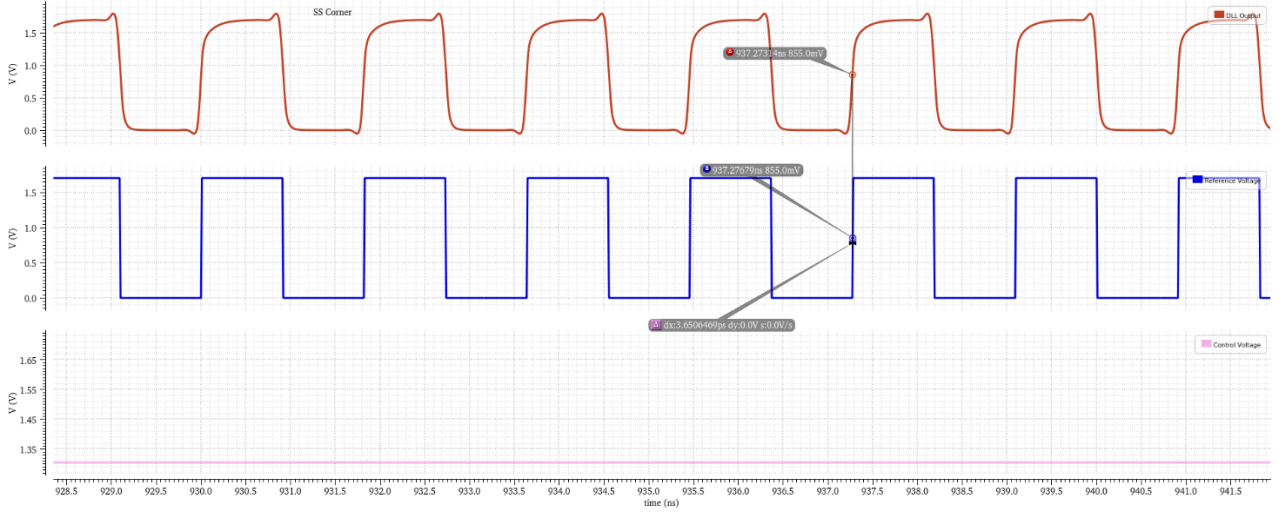
Plot 4(a): Full transient response of the DLL at Fast (FF) corner showing control voltage V_c settling at around 0.4V



Plot 4(b): Zoomed-in view of reference clock and DLL output at Fast (FF) corner showing a static phase offset of 13.3 ps



Plot 5(a): Full transient response of the DLL at Slow (SS) corner showing control voltage V_c settling at around 1.305V



Plot 5(b): Zoomed-in view of reference clock and DLL output at Slow (SS) corner showing a static phase offset of 3.65 ps

- **Calculation of DLL Loop Bandwidth**

Formula used for Bandwidth Calculation:

$$f_{BW} = \frac{(I_{cp} \cdot |K_{dl}| \cdot f_{ref})}{C_{LF}}$$

Condition	Fast Corner (FF)	Slow Corner (SS)
Reference Frequency (f_{ref})	500 MHz	550 MHz
VCDL Gain (K_{dl})	985.93 ps/V	563.00 ps/V
CP Current (I_{cp})	20 uA	20 uA
Filter Cap (C_{LF})	10 pF	10 pF
Loop Bandwidth (f_{BW})	985.93 kHz	619.30 kHz

Table 1: Calculated DLL Loop Bandwidth for loop filter cap of 10pF

- **Static Phase Error (SPO)**

Condition	Fast Corner (FF)	Nominal (NN)	Slow Corner (SS)
Measured SPO	13.2975 ps	7.41 ps	3.65 ps

Table 2: Static Phase Error (SPO) for different corners

- **Power Consumption**

Condition	Fast Corner (FF)	Nominal (NN)	Slow Corner (SS)
Vdd	1.89V	1.8V	1.71V
I _{avg}	0.58 mA	1.253 mA	2.783 mA
Power	1.098 mW	2.255 mW	4.758 mW

Table 3: DLL Power Consumption

4. Design Methodology and Analysis

4.1 Voltage-Controlled Delay Line (VCDL) Design

- **Architecture:** The VCDL is implemented as a chain of 20 current-starved inverters .
- **Design Process:**
- The design began by calculating the required delay for the VCDL to support the target frequency range of 500 MHz ($T = 2.0$ ns) to 550 MHz ($T = 1.818$ ns) .
- For each stage, the delay was optimized to ensure that the low-to-high (t_{PLH}) and high-to-low (t_{PHL}) propagation delays were approximately equal, thereby maintaining a 50% duty cycle .
- To ensure the 20-stage chain could achieve the desired delay across all process corners (Typical, Fast, and Slow), the transistors were sized accordingly.
- Extensive simulations across different corners confirmed that the VCDL meets the required 1.818 ns to 2.0 ns delay within the specified V_C range of 0.4V to 1.4V .

4.2 Phase Frequency Detector (PFD) Design

- **Logic Implementation:** The PFD was constructed using two D-Flip-Flops (DFF) and an AND gate .
- **Customization:** The DFFs were implemented using Verilog-A, allowing for the inclusion of an asynchronous reset signal.
- **Operational Refinements:** An additional DFF was added to the D-input of the reference clock flip-flop to skip the first edge of the reference clock. This prevents the system from attempting to lock onto unstable signals during the initial power-up phase.
- **Dead Zone Elimination:** In theory, a PFD can have a "dead zone" where it fails to detect very small phase differences. To prevent this and minimize the Static Phase Offset (SPO), a specific delay (t_{del}) was tuned within the AND gate . This ensures that the UP and DN pulses remain active long enough for the charge pump to respond even when the clocks are nearly aligned.

4.3 Charge Pump (CP) Design

- **Topology:** A source switched charge pump was selected to reduce non-ideal effects such as clock feedthrough and charge injection .
- **Sizing (gm/ID Method):** The gm/ID method was employed to size the charge pump transistors for the required 20 μA current.
- **Optimization for Saturation:** To ensure a small overdrive voltage drop across the two NMOS and two PMOS transistors, I_D/W values were evaluated for large channel lengths. The width W was calculated based on $I_D = 20 \mu\text{A}$
- **Op-Amp Integration:** An ideal Op-Amp (from the ahdLib) was utilized for biasing the PMOS transistors . The Op-Amp acts to keep the drain voltages of the NMOS and PMOS current sources equal, which significantly improves current matching and is essential for achieving a low SPO .
- **Loop Filter and Initialization:** A 10 pF capacitor was used as the loop filter to integrate the charge pump current into the control voltage (V_c) . To ensure the system starts from a known state and to speed up the locking process, a PMOS switch was included to initialize the capacitor charge to 1.8V upon reset.

4.4 Analysis of Results and Deviations from Theory

- **Static Phase Offset (SPO):** While theoretical models often assume a zero-offset lock, real-world factors like parasitic capacitance and finite reset delays introduce a small offset. The design achieved an SPO within the specified 25 ps limit across all tested corners.